

SEMINAR
CSE 3200

How to Write a Research Paper

A short, thick orange horizontal line is positioned below the title.

What is a Research Paper?

Understanding the Purpose & Types

A research paper is a formal, structured piece of academic writing that presents original analysis, argument, or findings on a specific topic, supported by evidence and properly cited sources.

Analytical

Breaks down an issue and evaluates it from multiple angles.

Argumentative

Takes a stance and defends it using evidence and logic.

Expository

Explains a topic clearly and thoroughly without bias.

Review Paper

Summarises existing literature on a subject.

Structure of a Research Paper

The 8 Core Sections — A–Z Roadmap

1

Title & Abstract

2

Introduction

3

Literature Review

4

Methodology

5

Results

6

Discussion

7

Conclusion

8

References

Step 0: Choosing Your Research Topic

Before You Write Anything

Pick What Interests You

Passion drives persistence. Choose a topic you genuinely want to explore.

Narrow It Down

Avoid overly broad topics. 'Climate Change' → 'Impact of CO₂ on coral reefs in 2015–2023'.

Check Feasibility

Ensure enough sources exist. Use Google Scholar or PubMed to gauge literature.

Form a Research Question

Every paper answers a clear question. Write it out before you begin.

Check for Gaps

Read recent papers to find what's unexplored — that's your research contribution.

Get Approval (if needed)

In academic settings, confirm your topic with your supervisor or course instructor.

Section 1: Title & Abstract

The Reader's First Impression

The Title

- Be specific and descriptive
- Reflect the paper's key variables
- Aim for 10–15 words
- Include the research context

The Abstract (150–300 words)

1. Background / Context
2. Research Problem or Objective
3. Methodology used
4. Key Results / Findings
5. Conclusion / Implication

✦ *Write this LAST — after completing the full paper.*

Section 2: Introduction

Set the Stage — Hook the Reader

The introduction is your paper's opening argument. It moves from broad context → specific problem → your study.

1

Hook / Opening Statement

Start with a striking fact, statistic, or question that captures attention.

2

Background Context

Briefly explain the broader topic and why it matters to the field.

3

Research Problem / Gap

Identify what is missing or unresolved in existing literature.

4

Research Objective & Questions

Clearly state what your paper aims to achieve and what questions it answers.

5

Thesis Statement

End with a concise statement of your argument or hypothesis.

Section 3: Literature Review

Situate Your Work in Existing Knowledge

The literature review shows you understand what has already been done and where your study fits.

How to Search

- Google Scholar
- PubMed / Scopus / JSTOR
- University Library Databases
- ResearchGate
- Search by keywords + year filter

How to Organize

- Group by theme, NOT just list sources
- Use a synthesis matrix or table
- Move from general → specific
- Identify agreements & contradictions
- Highlight the research gap

Common Mistakes

- Simply summarizing each paper
- Including irrelevant sources
- Not citing properly
- Ignoring recent literature (< 5 yrs)
- Missing key seminal works

Section 4: Methodology

How Did You Conduct Your Research?

The methodology section explains your research design so others can understand, evaluate, or replicate your study.

Research Design

Qualitative, Quantitative, or Mixed Methods? Experimental, descriptive, case study?

Sampling

Who/what is your sample? How was it selected? How large is it? Why?

Data Analysis

T Content analysis, statistical tests — explain your analytical approach.

Data Collection

How was data gathered? Surveys, interviews, experiments, observations, secondary data?

Instruments & Tools

What tools, software, or instruments were used? (e.g., SPSS, questionnaires, lab equipment)

Ethical Considerations

Informed consent, confidentiality, approval (if applicable).

Section 5: Results / Findings

What Did You Find?

Present your findings objectively — NO interpretation yet. Use visuals to enhance clarity.

DO

- Report findings clearly and logically
- Use tables, figures, charts for data
- Label all figures with numbers & captions
- Include statistical significance (p-values)
- Match results to your research questions
- Be precise — use exact numbers

DON'T

- Interpret or explain why results occurred
- Include raw data dumps
- Repeat all data in text if shown in table
- Add new information not in your data
- Use vague terms like 'many' or 'some'
- Hide or omit unfavourable results

Section 6: Discussion

Making Sense of Your Findings

This is where your voice matters most. Interpret, explain, and connect your results to the broader field.

A	Interpret Results:	Explain what your findings mean. Why did you get these results?
B	Connect to Literature:	Compare with previous studies — do you confirm, contradict, or extend them?
C	Address Research Questions:	Go back to your original questions. Did you answer them? How?
D	Acknowledge Limitations:	Be honest about weaknesses in methodology, sample size, or scope.
E	Implications:	What are the practical or theoretical implications of your findings?

Section 7: Conclusion

End Strong — Leave a Lasting Impression

The conclusion ties everything together. It should be concise but complete — no new information.

1. Restate the Research Problem

Briefly remind the reader of what you set out to investigate.

2. Summarise Key Findings

Highlight the most important results — not all of them, just the essentials.

3. Answer the Research Question

Directly state whether (and how) your research question was answered.

4. State Significance

Explain the contribution your research makes to the field.

5. Recommendations / Future Work

Suggest what should be explored next or what actions should be taken.

Section 8: References & Citations

Give Credit — Avoid Plagiarism

APA Style

Used in: Social Sciences, Psychology, Education

Example:

Smith, J. (2021). Research Methods. Oxford Press.

MLA Style

Used in: Humanities, Literature, Arts

Example:

Smith, John. "Research Methods." Oxford Press, 2021.

Chicago Style

Used in: History, Fine Arts, Business

Example:

Smith, John. Research Methods. Oxford: Oxford Press, 2021.

Citation Best Practices:

- Use citation managers: Zotero, Mendeley, EndNote
- Include only sources cited in the body of the paper
- Cite every source you paraphrase or quote
- Be consistent — pick one style and stick to it

Writing Tips & Academic Style

How to Write Like a Scholar

Use Formal Language

Avoid contractions (don't → do not), slang, and first person unless required.

Be Concise & Clear

Every sentence must serve a purpose. Cut redundancy ruthlessly.

Use Transition Words

Guide the reader: 'Furthermore...', 'In contrast...', 'This suggests that...'

Write Multiple Drafts

First draft = getting ideas out. Second = structure. Third = polishing.

Proofread Carefully

Check grammar, spelling, flow, citation format, and formatting consistency.

Use Active Voice

Prefer: 'We collected data' over 'Data was collected by us'.